**6. Hinge region**

Hinge region is the portion of heavy chain between the CH1 and CH2 domains. It is highly flexible and allows for movement of the Fab arms in relation to each other. The S values (sedimentation coefficient that is expressed in Svedberg units(s)) of immunoglobulins range from 7S- 19S.

### 13.6.2 Immunoglobulin Function

There are three major effector functions that enable antibodies to remove antigens and kill pathogens. Opsonization promotes antigen phagocytosis by macrophages and neutrophils. Complement activation by IgM and IgG can activate a pathway that leads to the generation of a collection of proteins that can perforate cell membranes. Antibody-dependent cell-mediated cytotoxicity (ADCC) can cause NK cell mediated death of target cells when antibody bound to the target cells associates with Fc receptors of natural killer (NK) cells.

**HOTS**

Which antibody protects the new born for few months against infections?

### 13.6.3 Properties and Activities of Immunoglobulin Classes

Each immunoglobulin class differs in its general properties, distribution in the body and interaction with other components of the host defensive systems.

**i) IgG**

1. IgG is the major immunoglobulin in human serum, accounting for 80% of the immunoglobulin pool.
2. It is present in blood plasma and tissue fluids. It has a monomeric structure.
3. IgG class acts against bacteria and viruses by opsonizing the invaders and neutralizing toxins and viruses.
4. IgG molecules are capable of fixing complement, except for IgG4.
5. It is the major antibody in the secondary immune response and it has half life of 23 days.
6. IgG is the only immunoglobulin molecule able to cross the placenta and provides natural immunity in utero and to the neonate at birth.

**ii) IgA**

It is present in the serum and in various bodily secretions and thus takes two forms – serum IgA and secretory IgA (sIgA)

**A) Serum IgA**

1. It accounts for about 12% of serum immunoglobulin.
2. In humans, over 80% of serum IgA exists in a monomeric form and the remaining existing as polymers in the form of dimers, trimers or tetramers. In polymeric IgA, the monomeric units are linked by disulphide bonds and joining (J) chain.
3. Serum IgA fixes complement via the alternative pathway. It has a half life of 5 days.

**B) Secretory IgA**

1. Secretory IgA is the primary immunoglobulin of mucosal associated lymphoid tissue (MALT). It is also found in saliva, tears, and breast milk.



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